HELP2PASS

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CCIE Security v6.0
Real Labs
Design Module Scenario 1.1 - FABD2 Inc.

Lab Workbook

3. One will be provided with free updates up to 120 days from the date of purchase, post that one need to renew his/her account to access the latest update. However one will continue to have access to their existing workbooks. If you pass the lab within 120 days, you are not eligible for further updates.

4. If one wish to renew their subscription/account, you need to renew within 120 days or before the account gets expired. Post 120 days one can renew their account however the renewal will be considered has a new purchase. Hence we encourage one to renew within 120 days of the purchase.

5. The renewal cost is 999 USD if one pay within 120 days, if one fail to renew then the cost will be equivalent to a new purchase. (The renewal price can be changed at any time, without informing the client)

6. Every workbook is uniquely identified for each user with hidden words. If one shares his/her workbooks with others, and if the system detects the share, the account will be banned and we will not entertain any explanation of any sort.

7. For any queries regarding Questions/Solutions, you can contact us on email: support@chinesedumps.com or skype @ chinesexams@gmail.com. Response time to any of the queries is 24 hours.

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9. The workbooks are in secured pdc format and delivered via email within 24 hours after payment is received.

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11. We do support devices running Windows OS, Mac OS, Android and Mac iOS only

12. We do not provide Refund in any circumstances once the product is sold.

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14. If there is any update, one will receive the update automatically on their registered email id.

15. Design Module will be given only 3 days before the CCIE exam

16. For any future update you can check our 'updates' page.

17. Labs are always published in phases. For e.g. if there is a new lab we publish it as First, Second, Third ... till Final release.

18. Client who have purchased our worbooks and services and wishes to attempt the lab, need to consult our experts before their CCIE Lab.

CCIE Design Guidelines

Before starting, please read the below guidelines:

1. In this module, you will be creating, analyzing, validating and optimizing a low-level network design. All relevant resources needed to successfully complete this module are provided within this module.
2. The menu bar on the main screen can be used to navigate to:

a) **Exam content.** Here you will find the exam questions. This module is scenario-based and contains about 30 to 35 web-based items. No device access is provided.

b) **Resources.** Here you will find provided resources. An initial set of resources is provided at the start of the module. Additional resources are provided as you progress. Resources are cumulative and remain available throughout the remainder of the module. It is recommended to read all the provided resources prior to answering a question.

c) **Guidelines.** If you want to review this guideline again during your exam, you can find them here.

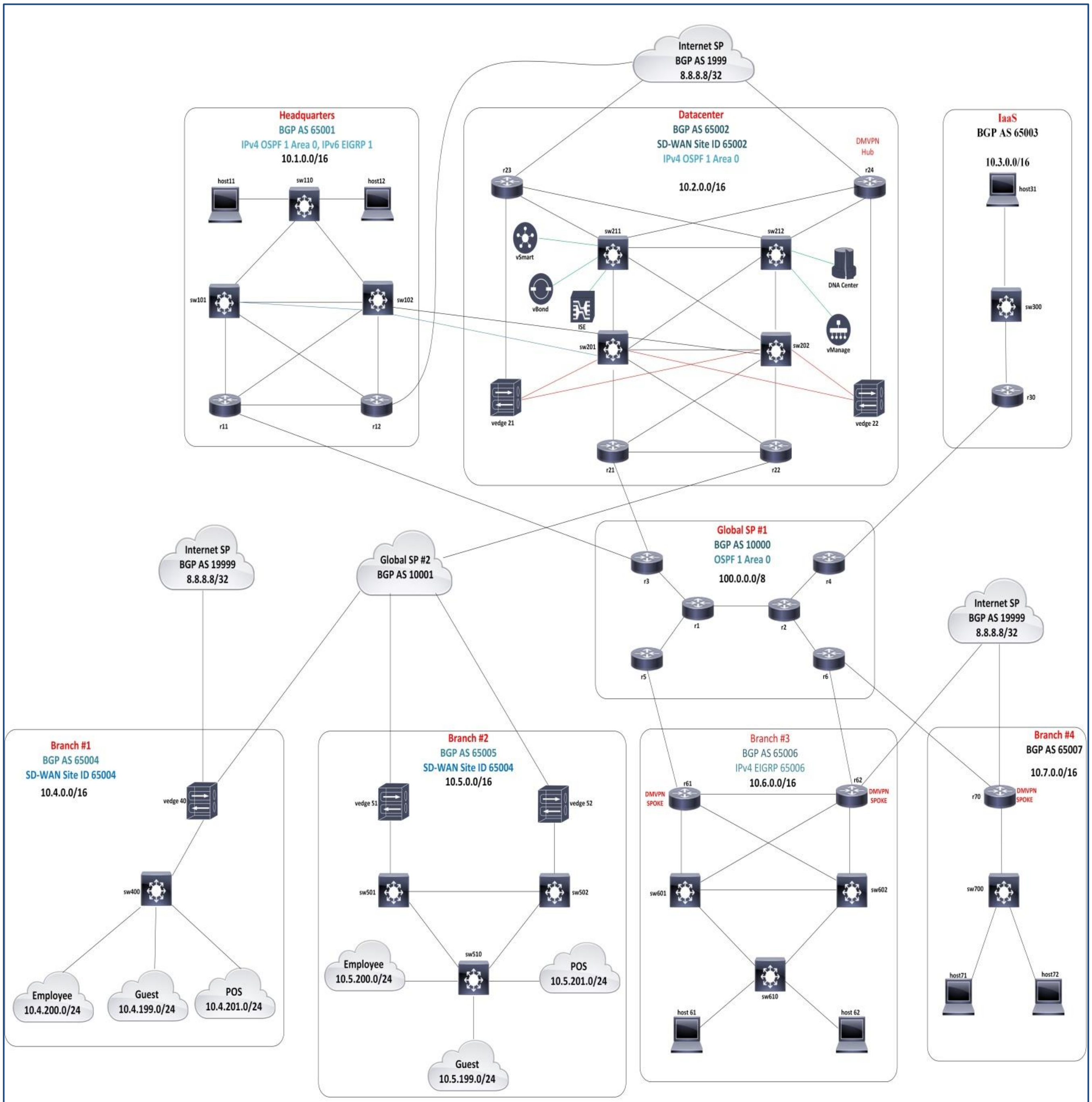
d) **Help.** Here you will find more information about the exam environment and functionalities.

e) **End Exam Section.** Clicking this button will end this exam section

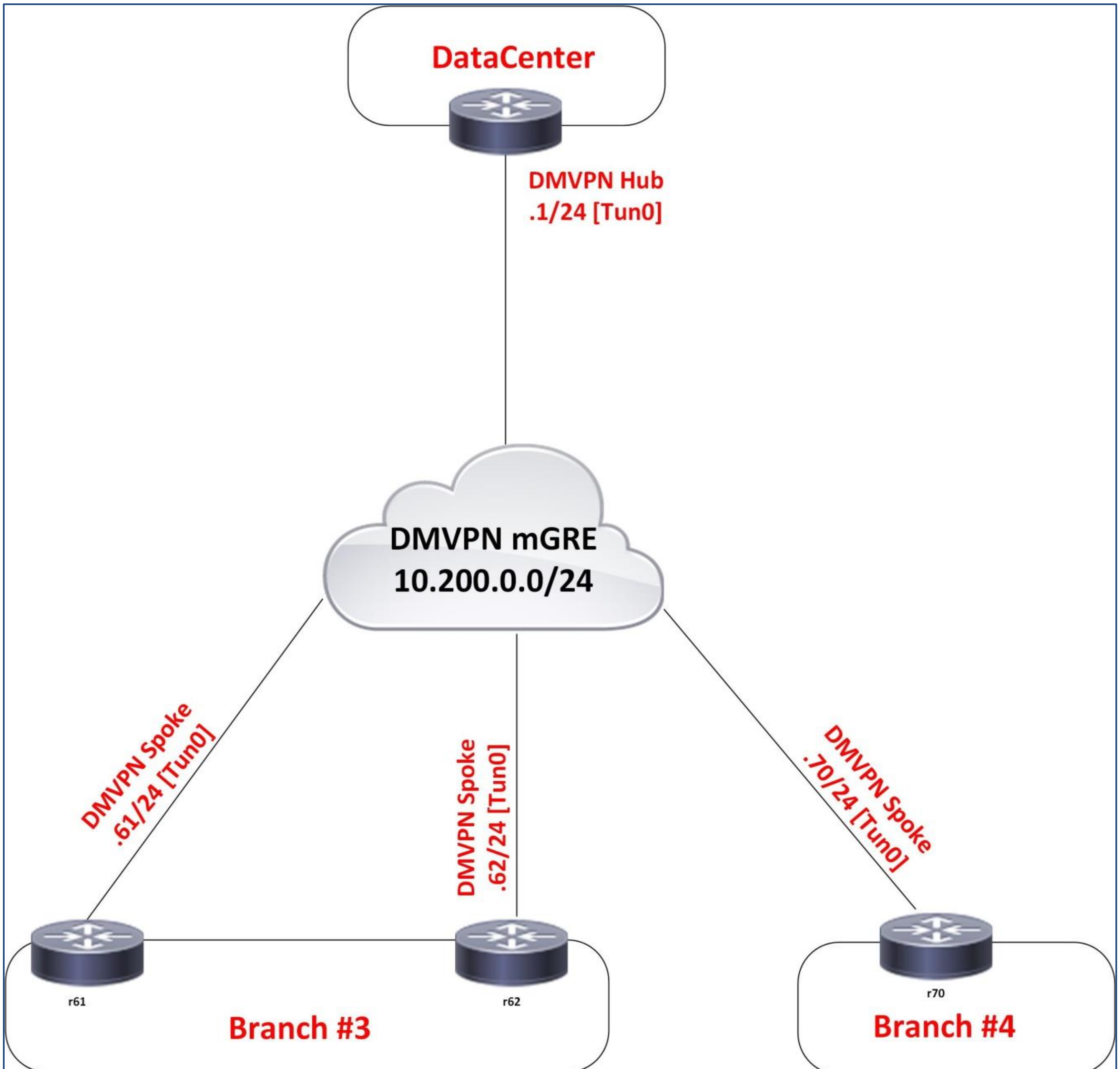
3. Backward navigation in this module is disabled; once you proceed to the next question, you will not be able to return to the previous question
4. Question point values will not be visible in this module. More complex items may have partial scoring opportunities
5. Item level feedback can be provided at question level. Feedback will be processed, but Cisco will not reach out to you to discuss any feedback provided. Any time spent on providing feedback will not be compensated.
6. Access to selected Cisco online documentation is available from your desktop. Access to select 3rd party product documentation (such as python) is available from the resources window under the “External Documentation” category.
7. If you suspect an issue with your exam environment, contact the lab proctor as soon as possible.
8. You have 3 hours to complete this module. If you finish early, you may start with the next module but any unused time will not be carried over to the next module.

Diagrams

Main Topology


Internet SP
BGP AS 1999
8.8.8.8/32
Headquarters
BGP AS 65001
IPv4 OSPF 1 Area 0, IPv6 EIGRP 1
10.1.0.0/16
host11
sw110
host12
sw101
sw102
r11
r12
Datacenter
BGP AS 65002
SD-WAN Site ID 65002
IPv4 OSPF 1 Area 0
10.2.0.0/16
DMVPN Hub
r23
r24
sw211
sw212
vSmart
vBond
ISE
DNA Center
vManage
sw201
sw202
vedge 21
vedge 22
r21
r22
IaaS
BGP AS 65003
10.3.0.0/16
host31
sw300
r30
Internet SP
BGP AS 19999
8.8.8.8/32
Global SP #2
BGP AS 10001
Global SP #1
BGP AS 10000
OSPF 1 Area 0
100.0.0.0/8
r3
r4
r1
r2
r5
r6
Internet SP
BGP AS 19999
8.8.8.8/32
Branch #1
BGP AS 65004
SD-WAN Site ID 65004
10.4.0.0/16
vedge 40
sw400
Employee
10.4.200.0/24
Guest
10.4.199.0/24
POS
10.4.201.0/24
Branch #2
BGP AS 65005
SD-WAN Site ID 65004
10.5.0.0/16
vedge 51
vedge 52
sw501
sw502
sw510
Employee
10.5.200.0/24
POS
10.5.201.0/24
Guest
10.5.199.0/24
Branch #3
BGP AS 65006
IPv4 EIGRP 65006
10.6.0.0/16
DMVPN SPOKE
r61
r62
DMVPN SPOKE
sw601
sw602
sw610
host 61
host 62
Branch #4
BGP AS 65007
10.7.0.0/16
r70
DMVPN SPOKE
sw700
host71
host72

DMVPN Topology



Background Information

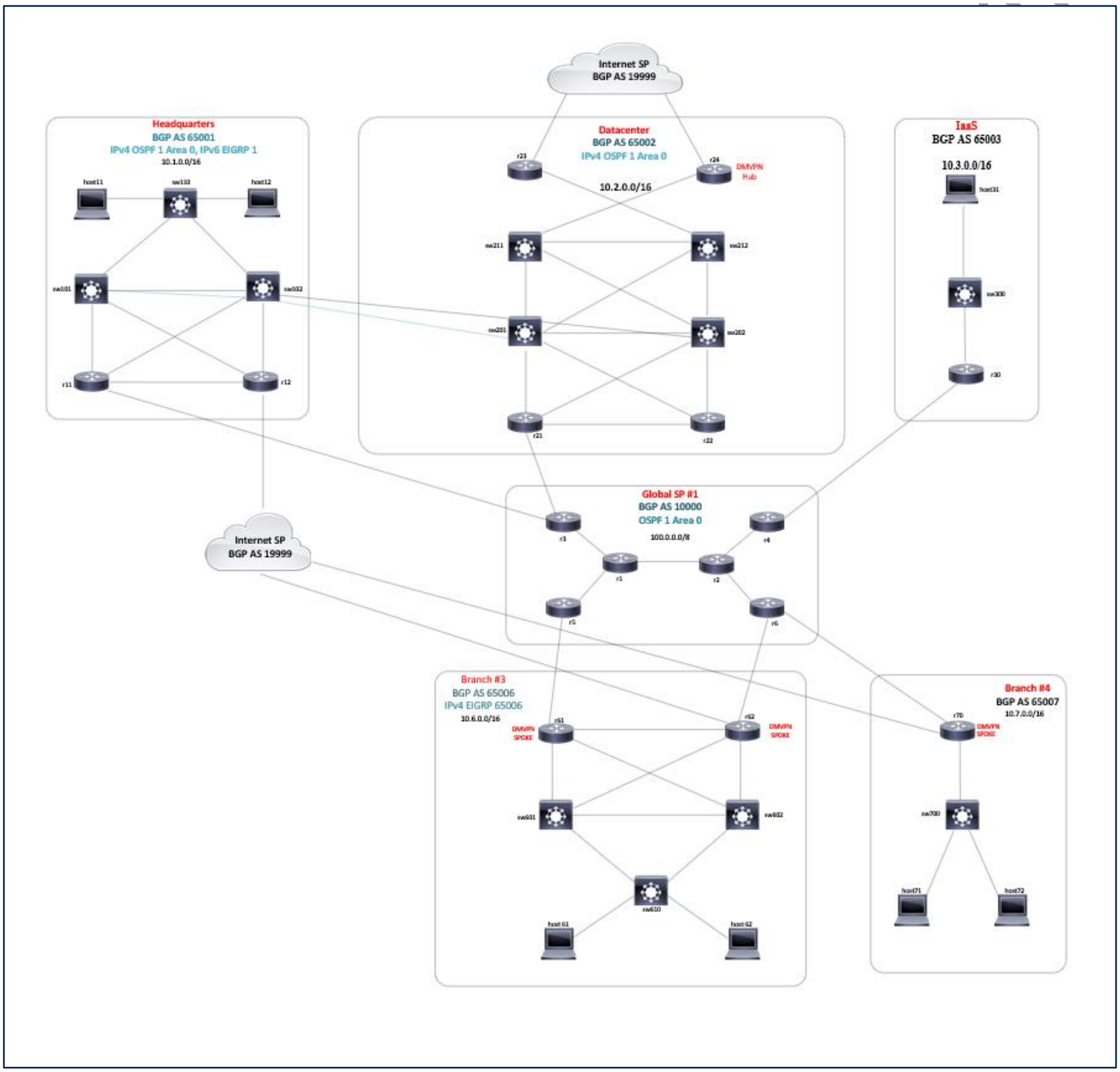
Introduction

FABD2 Inc. is a large national retailer with its headquarters and datacenter located in San Jose. The company has a large presence with more than 300 stores and 15 regional offices, all considered as branches. Like many other retailers, FABD2 is looking into making the customer experience even better by providing smartphone apps helping their customers navigating the stores even locating the items on the shelves. They also want to improve the self-service experience, both from a check-out perspective and by providing the customer with relevant offers during their store visit. At the same time, FABD2 is faced with requirements to cut the costs of operating and maintaining their networks.

FABD2 network has gone over a set of iterations and growth, and for the most part adheres to the classic paradigm of access, distribution, and core layer with Layer3 boundaries usually placed at the distribution layer switches. There are signs that the network has, from time to time, grown "organically and some of the design choices are less than optimal.

Current Environment Overview

The current install base consists of somewhat aging equipment built before DevOps and SecOps became a thing. The following drawing shows the high-level topology of the FABD2 network.



The network at the HQ location consists of one access switch sw110, two distribution/collapsed core switches sw101 and sw102, and two WAN edge routers r11 and r12. Router r11 connects to an MPLS L3VPN service provider (SP #1) while router r12 connects to an ISP. The switches sw101 and sw102 also have direct routed links to switches sw201 and sw202 in the DC.

The DC network follows the routed access approach, with switches sw211 and sw212 being the access switches, and sw201 and sw202 being the aggregation switches. There are four WAN edge routers in the DC Router r21 connects to SP #1. Routers r23 and r24 are dedicated for internet access, and r24 is also the DMVPN hub. Router r22 is not currently connected to any outside provider.

Branch networks, depending on their size, may either have just a single access switch and a router-on-stick that also acts as a WAN edge router or they may deploy a classic access / collapsed core architecture, and one or two WAN edge routers. All branches are connected to SP #1. Only a handful of branches also have a direct connection to the ISP. Most branches depend on transiting through DMVPN into DC for their internet access. All branches are members of the same DMVPN.

FABD2 is looking into modernizing the network, improving the existing design, and even introduce new technologies to bring the network into the next era. FABD2 is ultimately looking into lowering into the operational expenses but that must never occur at the cost of sacrificing current functionality. Whatever changes and additions to the network are performed, the current infrastructure must, remain stable until the new or improved one is in place. Implementing a new infrastructure will be handled in a phased approach in order to avoid making changes to the entire network at the same time. FABD2 management expects this to take 1-2 years, so careful and efficient planning is needed to be successful.

FABD2 hired an external partner to assess the current network status and provide their suggestions on how to bring the FABD2 network to the next level. The resulting report was recently handed over to FABD2 management, and below is an unsorted list of the suggested initiatives.

- **Preparing for the future**. Purpose: Migrate traditional enterprise campus networking to a software defined model. Aside from lowering operational costs, it also brings in the next-generation networking and security features.
- **SD-WAN**. Purpose: Lower costs of WAN circuits by downsizing existing MPLS circuits or even decommission them when it makes sense. Traffic to be offloaded to the internet whenever possible.
- **Optimization of current infrastructure**. Purpose: The assessment found several cases of less-than-optimal implementations of various technologies in nearly all areas of the current infrastructure.

You, as a promising CCIE candidate, has been brought in to work with relevant FABD2 teams to ensure these projects gets designed and implemented in the optimal way.

Cast and Characters

- **Travis Henderson:** FABD2 Network Manager
- **Hugh Donovan:** FABD2 Server Team
- **Anna Vasquez:** FABD2 RP/WAN team
- **Andrew Dunn-Lewis:** Procurement Manager, Communication

External Documents

- Python 3.8.3
  https://docs.python.org/3.8/
- Request 2.22
  https://requests.readthedocs.io/en/master/
  https://2.python-requests.org/en/v2.7.0/
- NCC client 0.4.4 Documentation
  https://ncclient.readthedocs.io/en/latest/

Emails:

Loop Guard and UDLD

From: "Travis Henderson" <travhend@fabd2inc.com>
To: "CCIE Candidate" <ccie2be@fabd2inc.com>
Subject: Loop Guard and UDLD

Hi there,

Sorry for the distraction with the EIGRP flaps earlier but it required our immediate attention. Thanks for helping there.

One more thing with STP... From what I know, STP is quite fragile and if BPDUs get lost in one direction on a link while the other one still works, there is still a risk of a switching loop. I'd like to avoid that. I've heard about STP Loop Guard and UDLD as a protection against similar issues but I am not acquainted with their details.

I'd appreciate if you could do a concise comparison of the UDLD and Loop Guard for us. Based on this, we will decide which one to deploy.

Travis

Refer to the new resource(s) available.

Question1: Fix every protection mechanism, indicate whether the individual statements are true, If any. (Select all that apply.)

	Mechanism	
Statements	**STP Loop Guard**	**UDLD**
Is STP-instance-aware	☐	☐
Must be enabled on both switches on a link to operate	☐	☐
Is STP-instance-agnostic	☐	☐
On detecting a fault, err-disables the port	☐	☐
On detecting a fault, blocks the port and declares an inconsistency	☐	☐
Protects against STP failures caused by absent BPDUs	☐	☐
Protects against native VLAN mismatches	☐	☐
Protects against STP failures caused by unidirectional links	☐	☐
Protects against faults caused by split fiber scenarios	☐	☐
Relies purely on existing messaging between switches	☐	☐

RE: Loop Guard and UDLD

From: "Travis Henderson" <travhend@fabd2inc.com>
To: "CCIE Candidate" <ccie2be@fabd2inc.com>
Subject: RE: Loop Guard and UDLD

Hi,
Thank you for the review, I'm still wondering about which mechanism to pick, though I've read somewhere that neither UDLD nor Loop Guard are perfect on their own. Can I run them in parallel? Alternatively, should I run a different combination of STP protective features? At the end of the day, I want to protect my switched network in HQ against issue arising from unidirectional link failures.

Travis

Refer to the new resource(s) available:

Question2: Based on the email from Travis, which two statements about the combination of the diverse mechanism are true? (Choose two)

- UDLD can be combined with Loop Guard
- Loop Guard can be combined with Root Guard
- UDLD and BPDU filter are mutually exclusive
- Loop Guard and STP Dispute are mutually exclusive
- Loop Guard and BPDU Guard are mutually exclusive

Answer:

- UDLD can be combined with Loop Guard
- UDLD and BPDU filter are mutually exclusive

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